

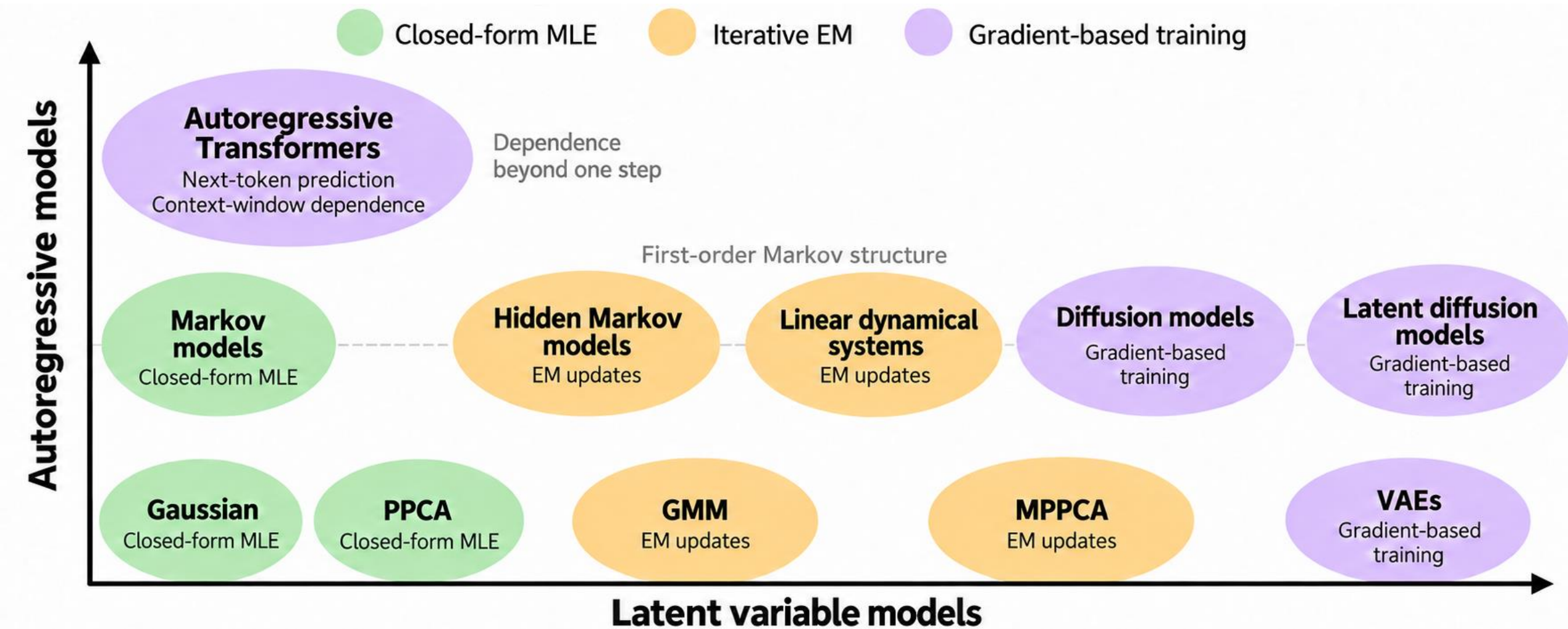
Deep Generative Models: Markov Models

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René Vidal

Director of the Center for Innovation in Data Engineering and Science (IDEAS),
Rachleff University Professor, University of Pennsylvania
Amazon Scholar & Chief Scientist at NORCE

Taxonomy of Generative Models



Lecture Outline

- Stochastic Processes
 - Definition and Examples
- Markov Models and Markov Chains
 - Definition
 - Transition Matrix
 - Examples
- Inference via Matrix Multiplication
- Learning via Maximum Likelihood

Stochastic Process

- **Definition:** A *stochastic process* $(X_0, X_1, \dots, X_T)$ is a sequence of random variables where each X_t takes values in the same sample space Ω (state space)
- **Example** (Bernoulli Process): X_t has 2 states and $\Omega := \{0, 1\}$

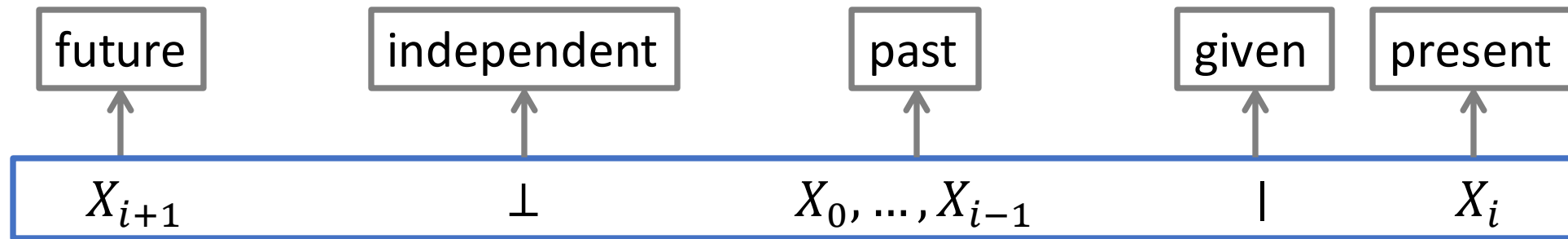
$$X_t \sim \text{Bernoulli}(q), \quad t = 0, \dots, T$$

- **Example** (Categorical Process): X_t has K states and $\Omega := \{1, \dots, K\}$

$$X_t \sim \text{Cat}(\pi), \quad t = 0, \dots, T$$

Markov Property (circa 1906)

- **Issue:** Modeling the joint distribution $p(X_0, X_1, \dots, X_T)$ might require **exponentially many parameters** in the absence of any assumptions.
- **Conditional Independence Assumption** (Markov property):



- **Consequence:** We now only need **linearly many parameters**:
$$p(x_0, \dots, x_T) = p(x_0)p(x_1 | x_0)p(x_2 | \cancel{x_0}, x_1) \cdots p(x_T | \cancel{x_0}, \dots, \cancel{x_{T-2}}, x_{T-1})$$
$$= p(x_0)p(x_1 | x_0)p(x_2 | x_1) \cdots p(x_T | x_{T-1})$$

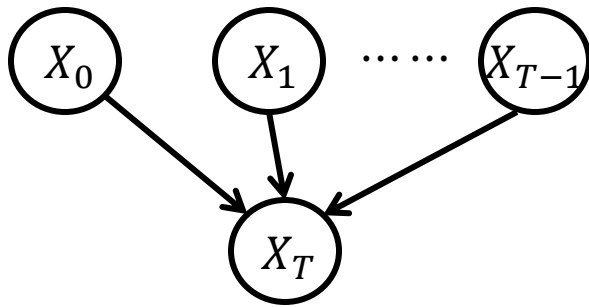
$\underbrace{\hspace{15em}}_{T + 1 \text{ terms}}$

Markov Chains

- **Definition:** A (discrete time) *Markov chain* is a stochastic process $(X_0, X_1, \dots, X_T)$ with the *Markov property*

$$p(x_0, \dots, x_T) = p(x_0)p(x_1 | x_0)p(x_2 | x_1) \cdots p(x_T | x_{T-1})$$

- Without the Markov Property:



- With the Markov Property:



Parameters of Markov Chains

- **Initial Probability:** π_i is the probability that X_0 starts at state i

$$\pi_i := p(X_0 = i) \quad \forall i \in \Omega = \{1, \dots, K\}$$

- **Transition Probability:** a_{ij} is the probability that X_t transitions from state i to j

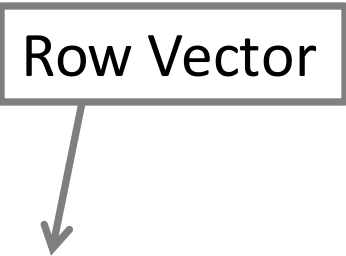
$$a_{ij} := p(X_{t+1} = j \mid X_t = i) \quad \forall i, j \in \Omega = \{1, \dots, K\}$$

- **Matrix and Vector Notations:**

$$\mathbf{A} := \begin{bmatrix} a_{11} & \dots & a_{1K} \\ \vdots & \ddots & \vdots \\ a_{K1} & \dots & a_{KK} \end{bmatrix} \in \mathbb{R}^{K \times K},$$

$$\boldsymbol{\pi} := [\pi_1, \dots, \pi_K] \in \mathbb{R}^{1 \times K}$$

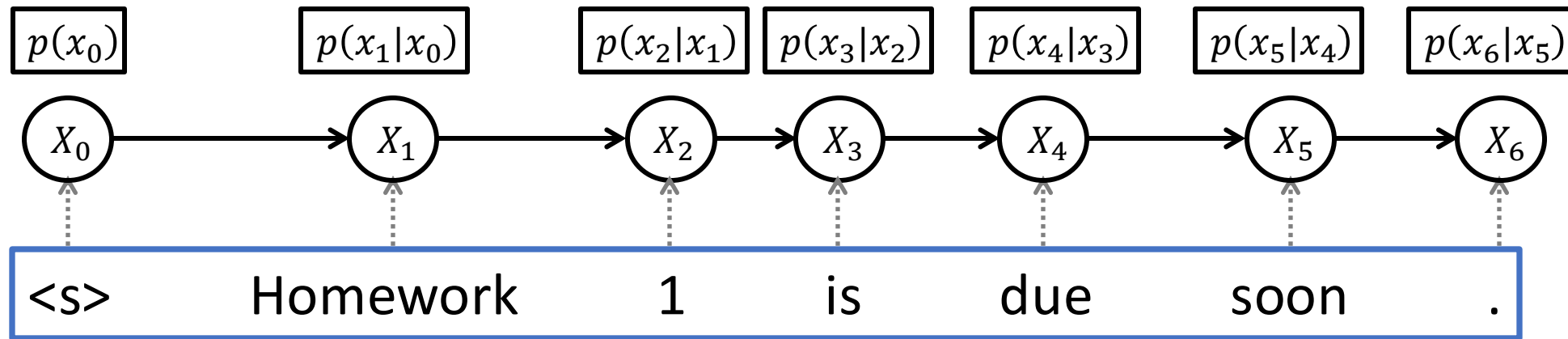
Row Vector



A Markov chain is fully specified by its parameters $\theta := (\boldsymbol{\pi}, \mathbf{A})$

Example: Markov Sentence Model

- State Space $\Omega = \{\text{all possible words}\}$
 - The following are viewed as words and included in the state space
 - $\langle s \rangle$: the start of the sentence
 - Digits
 - Punctuations
- Each sentence is a Markov chain where the words are random variables:



- Meaning of $p(x_{t+1} | x_t)$: Given that the current word is x_t , what is the probability that the next word is x_{t+1} ?

Example: DNA Sequencing

- State Space $\Omega = \{\mathcal{A}, \mathcal{C}, \mathcal{G}, \mathcal{T}\}$

- Transition Matrix A :

	$\mathcal{A}$	$\mathcal{C}$	$\mathcal{G}$	$\mathcal{T}$
$\mathcal{A}$	0.359	0.143	0.167	0.331
$\mathcal{C}$	0.384	0.156	0.023	0.437
$\mathcal{G}$	0.306	0.199	0.150	0.345
$\mathcal{T}$	0.284	0.182	0.177	0.357

Initial Probability Vector $\boldsymbol{\pi}$:

$\mathcal{A}$	0.25
$\mathcal{C}$	0.25
$\mathcal{G}$	0.25
$\mathcal{T}$	0.25

- **Question 1:** Given $\mathcal{C}$, what is the probability of getting DNA sequence $\mathcal{CTGAC}$?

- **Answer 1:**

$$p(\mathcal{CTGAC} \mid X_0 = \mathcal{C}) = p(\mathcal{T} \mid \mathcal{C}) \cdot p(\mathcal{G} \mid \mathcal{T}) \cdot p(\mathcal{A} \mid \mathcal{G}) \cdot p(\mathcal{C} \mid \mathcal{A}) \approx 0.00338$$



Example: DNA Sequencing

- State Space $\Omega = \{\mathcal{A}, \mathcal{C}, \mathcal{G}, \mathcal{T}\}$

	$\mathcal{A}$	$\mathcal{C}$	$\mathcal{G}$	$\mathcal{T}$	
$\mathcal{A}$	0.359	0.143	0.167	0.331	$\mathcal{A}$ 0.25
$\mathcal{C}$	0.384	0.156	0.023	0.437	$\mathcal{C}$ 0.25
$\mathcal{G}$	0.306	0.199	0.150	0.345	$\mathcal{G}$ 0.25
$\mathcal{T}$	0.284	0.182	0.177	0.357	$\mathcal{T}$ 0.25

- Question 2:** What's the probability of $X_2 = \mathcal{A}$ given $X_0 = \mathcal{C}$?

- Answer 2:** The state transition is $\mathcal{C} \rightarrow x_1 \rightarrow \mathcal{A}$ for all possible $x_1 \in \Omega$:

$$p(X_2 = \mathcal{A} \mid X_0 = \mathcal{C}) = \sum_{x_1 \in \Omega} p(X_2 = \mathcal{A} \mid X_1 = x_1) \cdot p(X_1 = x_1 \mid X_0 = \mathcal{C})$$

$$= [0.384, 0.156, 0.023, 0.437] \begin{bmatrix} 0.359 \\ 0.384 \\ 0.306 \\ 0.284 \end{bmatrix}$$

- This is the inner product of the second row and first column of the transition matrix A
- This is the (2,1)-th entry of A^2

Example: DNA Sequencing

- State Space $\Omega = \{\mathcal{A}, \mathcal{C}, \mathcal{G}, \mathcal{T}\}$

	$\mathcal{A}$	$\mathcal{C}$	$\mathcal{G}$	$\mathcal{T}$		
$\mathcal{A}$	0.359	0.143	0.167	0.331	$\mathcal{A}$	0.25
$\mathcal{C}$	0.384	0.156	0.023	0.437	$\mathcal{C}$	0.25
$\mathcal{G}$	0.306	0.199	0.150	0.345	$\mathcal{G}$	0.25
$\mathcal{T}$	0.284	0.182	0.177	0.357	$\mathcal{T}$	0.25

- Question 3: What's the probability of $X_2 = \mathcal{A}$?
- Answer 3: The state transition is $x_0 \rightarrow x_1 \rightarrow \mathcal{A}$ for all possible $x_0, x_1 \in \Omega$:

$$p(X_2 = \mathcal{A}) = \sum_{x_0 \in \Omega} p(X_2 = \mathcal{A} | X_0 = x_0) \cdot p(X_0 = x_0)$$

Question 2

Initial Probability

- This is the inner product of the vector π and the first column of $\mathbf{A}^2$: $\pi \mathbf{A}^2 \mathbf{e}_1$

Transition Matrix and Distribution of Future States

- State Space $\Omega = \{1, \dots, K\}$
- $(\mathbf{A}^s)_{ij}$: the (i, j) -th entry of $\mathbf{A}^s$
- $(\mathbf{A}^s)_{:j}$: the j -th column of $\mathbf{A}^s$
- $(\cdot)_j$: the j -th entry of a vector

Transition Matrix A and initial probability distribution π :

$$\mathbf{A} := \begin{bmatrix} a_{11} & \cdots & a_{1K} \\ \vdots & \ddots & \vdots \\ a_{K1} & \cdots & a_{KK} \end{bmatrix} \in \mathbb{R}^{K \times K}, \quad \boldsymbol{\pi} := [\pi_1, \dots, \pi_K] \in \mathbb{R}^{1 \times K}$$

- Claim 1: $p(X_{t+s} = j \mid X_t = i) = (\mathbf{A}^s)_{ij} \quad (\forall s, t, i, j)$
- Claim 2: $p(X_s = j) = (\boldsymbol{\pi} \mathbf{A}^s)_j \quad (\forall s, j)$
- Proof of Claim 2:

$$p(X_s = j) = \sum_{i \in \Omega} p(X_s = j \mid X_0 = i) \cdot p(X_0 = i) = \sum_{i \in \Omega} (\mathbf{A}^s)_{ij} \cdot \pi_i = \boldsymbol{\pi} (\mathbf{A}^s)_{:j} = (\boldsymbol{\pi} \mathbf{A}^s)_j$$

- Proof of Claim 1: By induction (next page)

Claim 1

Proof of Claim 1

- State Space $\Omega = \{1, \dots, K\}$
- $(\mathbf{A}^s)_{ij}$: the (i, j) -th entry of $\mathbf{A}^s$
- $(\mathbf{A}^s)_{:j}$: the j -th column of $\mathbf{A}^s$
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Transition Matrix A and initial probability distribution π :

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- **Claim 1**: $p(X_{t+s} = j \mid X_t = i) = (\mathbf{A}^s)_{ij}$
- **Proof of Claim 1 (Induction)**:
 - $\forall s, t$, it is easy to prove **shift invariance**: $p(X_{t+s} = j \mid X_t = i) = p(X_s = j \mid X_0 = i)$
 - Next we prove $p(X_s = j \mid X_0 = i) = \mathbf{A}_{ij}^s$ by induction on s :
 - The base case $s = 1$ follows from the definition of $\mathbf{A}$
 - Suppose we have $p(X_{s-1} = j \mid X_0 = i) = \mathbf{A}_{ij}^{s-1}$ then:
$$p(X_s = j \mid X_0 = i) = \sum_{k \in \Omega} p(X_s = j \mid X_{s-1} = k) \cdot p(X_{s-1} = k \mid X_0 = i) = \sum_{k \in \Omega} a_{kj} \cdot (\mathbf{A}^{s-1})_{ik} = (\mathbf{A}^s)_{ij}$$

Eigenvalues of Transition Matrix

$$\mathbf{A} := \begin{bmatrix} a_{11} & \cdots & a_{1K} \\ \vdots & \ddots & \vdots \\ a_{K1} & \cdots & a_{KK} \end{bmatrix} \in \mathbb{R}^{K \times K}$$

- **Proposition.** Let $\lambda_1, \dots, \lambda_K$ be eigenvalues of $\mathbf{A}$. Then

$$\max_{k=1, \dots, K} |\lambda_k| = 1 \quad \text{and} \quad \mathbf{A} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

- **Proof:** Let $(\lambda, \mathbf{u})$ be an eigen-pair, i.e., $\mathbf{A}\mathbf{u} = \lambda\mathbf{u}$, $\mathbf{u} = [u_1, \dots, u_K]^\top$ and $\|\mathbf{u}\|_2 = 1$. Let i^* be the index such that $|u_i|$ is maximized, i.e., $i^* = \operatorname{argmax}_i |u_i|$. Then, $\mathbf{A}\mathbf{u} = \lambda\mathbf{u}$ implies $\sum_j a_{i^*j} u_j = \lambda u_{i^*}$, which furthermore gives

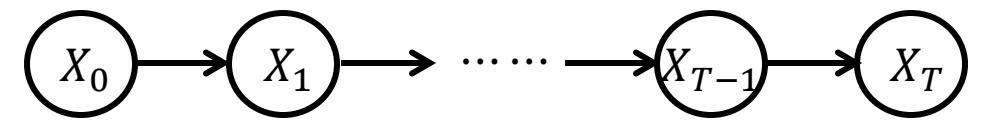
$$|\lambda| \leq \left| \frac{\sum_j a_{i^*j} u_j}{u_{i^*}} \right| \leq \sum_j |a_{i^*j}| \cdot \left| \frac{u_j}{u_{i^*}} \right| \leq \sum_j |a_{i^*j}| = \sum_j a_{i^*j} = 1.$$

Finally, since $\sum_j a_{ij} = 1$ for all $i = 1, \dots, K$, then $\mathbf{A}\mathbf{1} = \mathbf{1}$, hence $\max_{k=1, \dots, K} |\lambda_k| = 1$.

Learning the Parameters θ from Data

- We have derived some results based on the transition matrix ...
- In practice, we are given data samples rather than the transition matrix
- We will assume the data are sampled from a Markov chain, and then compute the transition matrix from data via **Maximum Likelihood Estimation (MLE)**

MLE of Markov Chains



- Assume we have N i.i.d. samples $\{\mathbf{x}^{(n)}\}_{n=1}^N$ from distribution $p_{\theta}(\mathbf{x})$
 - $\mathbf{x} := (x_0, \dots, x_T)$ each x_t can take on K different values
 - $\theta = (\mathbf{A}, \boldsymbol{\pi})$: unknown transition matrix and initial probability distribution

- MLE:

$$(\hat{\mathbf{A}}_{ML}, \hat{\boldsymbol{\pi}}_{ML}) = \operatorname{argmax}_{\mathbf{A}, \boldsymbol{\pi}} \prod_{n=1}^N p_{\mathbf{A}, \boldsymbol{\pi}}(\mathbf{x}^{(n)})$$

Markov Property

$$(\hat{\mathbf{A}}_{ML}, \hat{\boldsymbol{\pi}}_{ML}) = \operatorname{argmax}_{\mathbf{A}, \boldsymbol{\pi}} \prod_{n=1}^N p_{\boldsymbol{\pi}}(x_0^{(n)}) \prod_{t=1}^T p_{\mathbf{A}}(x_t^{(n)} | x_{t-1}^{(n)})$$

Variables are separable

Similar to
estimating
 $\hat{\mathbf{A}}_{ML}$, so left as
an exercise

$$\hat{\boldsymbol{\pi}}_{ML} = \operatorname{argmax}_{\boldsymbol{\pi}} \prod_{n=1}^N p_{\boldsymbol{\pi}}(x_0^{(n)})$$

Our focus next

$$\hat{\mathbf{A}}_{ML} = \operatorname{argmax}_{\mathbf{A}} \prod_{n=1}^N \prod_{t=1}^T p_{\mathbf{A}}(x_t^{(n)} | x_{t-1}^{(n)})$$

Simplifying The MLE

- $\mathbb{I}(\cdot)$: indicator function
- N_{ij} : the number of samples with transitions from state i to state j , i.e.,

$$N_{ij} := \sum_{n=1}^N \sum_{t=0}^T \mathbb{I} \left(x_t^{(n)} = j, x_{t-1}^{(n)} = i \right)$$

Then we have:

1. $p_A \left(x_t^{(n)} \mid x_{t-1}^{(n)} \right) = \prod_{i=1}^K \prod_{j=1}^K (a_{ij})^{\mathbb{I}(x_t^{(n)}=j, x_{t-1}^{(n)}=i)}$
2. $\prod_{n=1}^N \prod_{t=1}^T p_A \left(x_t^{(n)} \mid x_{t-1}^{(n)} \right) = \prod_{n=1}^N \prod_{t=1}^T \prod_{i=1}^K \prod_{j=1}^K (a_{ij})^{\mathbb{I}(x_t^{(n)}=j, x_{t-1}^{(n)}=i)}$
 $= \prod_{i=1}^K \prod_{j=1}^K (a_{ij})^{N_{ij}}$

This gives:

$$\hat{A}_{ML} = \operatorname{argmax}_A \prod_{i=1}^K \prod_{j=1}^K (a_{ij})^{N_{ij}}$$

$$\hat{A}_{ML} = \operatorname{argmax}_A \prod_{n=1}^N \prod_{t=1}^T p_A \left(x_t^{(n)} \mid x_{t-1}^{(n)} \right)$$

Simplifying The MLE

$$\hat{\mathbf{A}}_{ML} = \operatorname{argmax}_{\mathbf{A}} \prod_{i=1}^K \prod_{j=1}^K (a_{ij})^{N_{ij}}$$

- N_{ij} : the number of samples with transitions from state i to state j

Taking logarithm and adding constraints $\sum_j a_{ij} = 1$:

$$\hat{\mathbf{A}}_{ML} = \operatorname{argmax}_{\mathbf{A}} \sum_{i=1}^K \sum_{j=1}^K N_{ij} \log a_{ij} \quad \text{subject to} \quad \sum_{j=1}^K a_{ij} = 1 \quad (\forall i)$$

↓ Variables are separable

Solve the following for every $i = 1, \dots, K$:

$$\hat{a}_{ij_{ML}} = \operatorname{argmax}_{a_{ij}} \sum_{j=1}^K N_{ij} \log a_{ij} \quad \text{subject to} \quad \sum_{j=1}^K a_{ij} = 1$$

Remark: We have seen how to solve it using Lagrangian multipliers (recall *EM for Gaussian Mixture Models*)

$$\hat{a}_{ij_{ML}} = \frac{N_{ij}}{\sum_{j=1}^K N_{ij}}$$

Remark: The optimal transition matrix can be found by simply counting and classifying the number of the transitions of the sample states!

Conclusion

- Markov chains have several applications
 - Modeling text sequences
 - Modeling gene sequences
- The ML estimate $\hat{a}_{ij_{ML}}$ of the transition matrix is given by

$$\hat{a}_{ij_{ML}} = \frac{N_{ij}}{\sum_{j=1}^K N_{ij}}$$

Number of transitions from state i to j

- Similarly, the ML estimate $\hat{\pi}_{i_{ML}}$ of the initial probability is given as

$$\hat{\pi}_{i_{ML}} = \frac{N_{i0}}{\sum_{j=1}^K N_{j0}}$$

Number of times we start in state i